

Neural Networks Assignment 1

Summary Report

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Advanced Machine Learning

1. Introduction and Objective

The purpose of this assignment was to explore how different neural network design choices influence the performance of a sentiment classification model. I worked with the IMDB movie review dataset, where reviews are classified as positive or negative. The goal was to develop models that could learn from the training reviews and still perform well on reviews they had not seen before.

Instead of relying on one neural network, I tested several changes to the model. I compared network depth, increased the number of hidden units, changed the loss function, tested a different activation function, and added dropout regularization. My main focus was generalization. A model can perform very well on training data without performing equally well on new data, so validation accuracy was used to compare the experiments before the final model was evaluated on the untouched test set.

2. Approach and Experimental Procedure

I began with a relatively simple neural network and changed important settings one at a time. This made it easier to understand how each modeling decision affected performance. The first comparison examined network depth by comparing one hidden layer with three hidden layers. I then examined network width by testing 32 and 64 hidden units.

The next experiments focused on training choices rather than network size. Mean squared error (MSE) was tested instead of binary cross-entropy, and tanh was tested instead of ReLU. Finally, I added dropout regularization to examine whether strong validation performance could be maintained while reducing the risk of overfitting.

For each experiment, I monitored validation accuracy across training epochs and recorded the best validation result and the epoch where it occurred. I then compared the experiments as a group before selecting a configuration for final test evaluation.

3. Results and Model Comparison

The table below summarizes the strongest validation result from each experiment. The results were close overall, which made the comparison useful for understanding whether additional complexity actually produced a meaningful improvement.

Experiment	Best Validation Accuracy	Best Epoch
1 Hidden Layer	88.18%	7
3 Hidden Layers	88.45%	3
32 Hidden Units	88.61%	2
64 Hidden Units	88.71%	5
MSE Loss	88.71%	3
Tanh Activation	88.28%	3
Dropout	88.70%	3

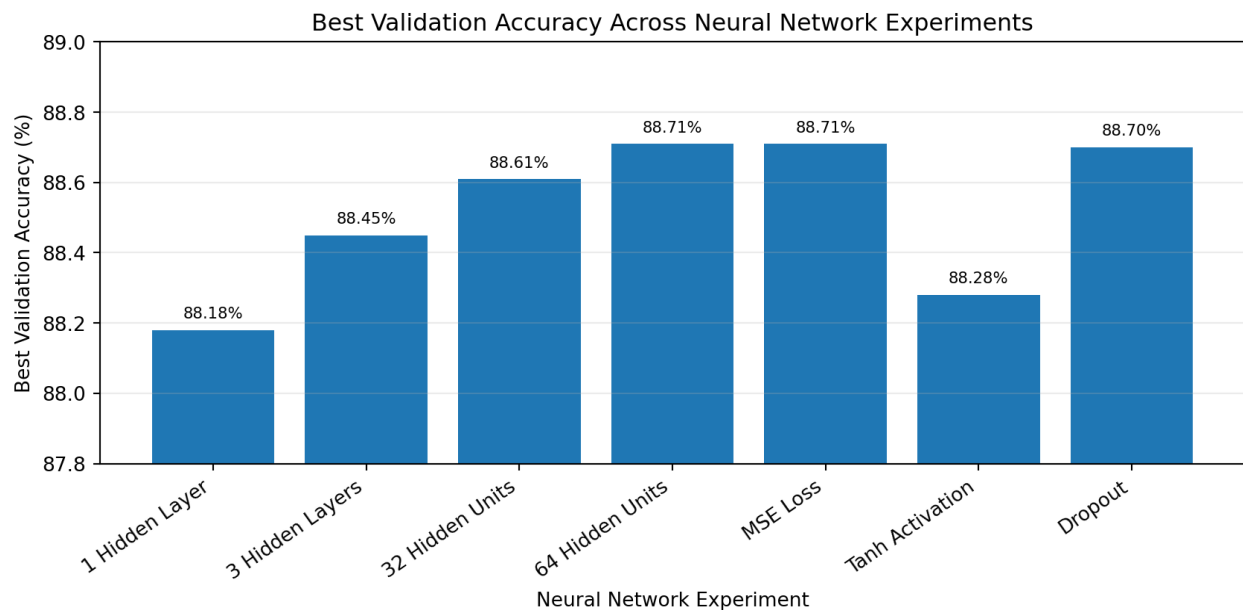


Figure 1. Best validation accuracy across the seven neural network experiments.

The lowest best validation accuracy was 88.18%, while the highest was 88.71%, a difference of only 0.53 percentage points. The 64-unit model and the MSE experiment produced the highest observed validation accuracy at 88.71%. The dropout model was almost identical at 88.70%, followed by the 32-unit model at 88.61%. These results showed that several different configurations were able to reach similar performance and that a more complicated model did not automatically create a large improvement.

4. Interpretation of the Experiments

Hidden Layers

The one-hidden-layer model reached a best validation accuracy of 88.18%, while the three-hidden-layer model reached 88.45%. Although the deeper model performed slightly better during validation, the test results showed a different pattern. The one-layer model achieved 87.14% test accuracy, compared with 86.18% for the three-layer model. This was important because it showed that adding depth did not automatically improve performance on unseen data.

Hidden Units

Increasing the hidden-layer size produced a small improvement. The 32-unit model achieved 88.61% validation accuracy, while the 64-unit model reached 88.71%. The improvement was only 0.10 percentage points. This suggests that the additional capacity helped, but the benefit was modest rather than dramatic.

Loss Function

Using MSE instead of binary cross-entropy produced a best validation accuracy of 88.71%, tying the 64-unit experiment for the highest observed validation result. I found this useful because it showed that alternative training settings can remain competitive. The result also reinforced the value of testing choices rather than assuming in advance how a particular change will affect performance.

Activation Function

The tanh model achieved a best validation accuracy of 88.28%. This was still a reasonable result, but it was below the stronger ReLU-based configurations. In my experiments, ReLU therefore gave better validation performance than tanh for this sentiment classification task.

Dropout and Overfitting

The dropout model achieved 88.70% validation accuracy, only 0.01 percentage points below the highest result. I also observed that training accuracy could continue increasing after validation performance stopped improving. This pattern suggests overfitting: the model becomes better at the training examples without gaining the same improvement on unseen data. Dropout was useful because it provided a way to control model complexity while maintaining strong validation performance.

5. Overall Findings

Across all experiments, no single change completely transformed the model's performance. Most configurations stayed within a narrow validation range. The comparison between one and three hidden layers was especially informative because the deeper model had slightly higher validation accuracy but lower test accuracy. The hidden-unit comparison showed a similar lesson: increasing capacity helped slightly, but the improvement was small.

6. Final Model Evaluation

After reviewing the experiments, I selected the 64-unit configuration for final evaluation. A fresh model was trained for five epochs and then evaluated using the untouched test dataset. The final model achieved 87.92% test accuracy with a test loss of 0.3085.

Final Test Accuracy: 87.92% Final Test Loss: 0.3085
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The final test accuracy was reasonably close to the validation performance observed during model development. This suggests that the selected configuration generalized well beyond the data used during training and validation. The result also supports an important lesson from the earlier experiments: a model does not need to become extremely deep to achieve strong performance. A carefully selected configuration can provide a useful balance between model capacity and generalization.

7. Conclusion and Recommendations

This assignment helped me understand that neural network tuning involves more than simply adding layers or increasing the number of units. Each modeling decision can affect training, validation, and test performance differently. A more complicated network does not necessarily perform better on unseen data.

Among my experiments, the 64-unit model and the MSE model achieved the highest validation accuracy of 88.71%, while the dropout model was extremely close at 88.70%. The final 64-unit model achieved 87.92% accuracy on the untouched test set with a loss of 0.3085. Based on the complete comparison, I would retain the 64-unit configuration as the final model because it provided one of the strongest validation results and also performed well on the test data.

If I continued this analysis, I would test different dropout rates and use early stopping. Early stopping could end training once validation performance stops improving, which may reduce unnecessary training and help control overfitting. I would compare these changes using the same validation process so that the models could be evaluated consistently.

My main takeaway is that neural network tuning is a process of finding a balance. Increasing model complexity can sometimes improve performance, but that improvement should be confirmed using validation and test data. In this assignment, careful comparison was more valuable than simply choosing the largest or deepest network.

Technical implementation and raw training outputs are intentionally excluded from this summary report. The accompanying Jupyter notebook contains the complete code and experiment outputs.